Nattanitcha Promboon

PRODUCT DESIGN & PROTOTYPING · CO-OP INTERN

nattanitchafarm@gmail.com
Portfolio

Full-time co-op · 13 Jan – 28 May 2027

EDUCATION

King Mongkut's University of Technology Thonburi (KMUTT)
B.Sc. Media Technology
2023 – present · co-op track
GPA 3.61 / 4.00

Wittayanukulnaree School
Science–Mathematics–Technology & Environment program

SKILLS

Design / Prototyping
Sketching & illustration · UI & motion design · CAD → 3D printing on my own printer · assembly & soldering · sewing & embroidery

Electronics
ESP32 · Arduino · Hall sensors · addressable LEDs · 18650 pack with BMS & USB-C charging

Tools
Figma · Tinkercad · SolidWorks (learning) · Blender · Maya · Photoshop · Illustrator · Creality Print · VS Code · Word · Excel · PowerPoint

AI workflow
Direct AI agents (Claude Code, Codex, Antigravity) with work orders that quote my requirements word for word; a separate reviewer checks each output; repeat runs and compare them, like checking a research instrument's reliability; log every step. Designer, not a programmer by background.

AWARDS & ACTIVITIES

1st Place — KMUTT Deep Sea Jam Hackathon 2026 (DeepCore)

Staff — FANTASTECH Media Exhibition 2026, KMUTT

Freelance — commission illustration & handmade dolls (ongoing)

LANGUAGES

Thai — native
English — working proficiency (reading & writing technical documents)

PROFILE

Year-4 Media Technology student who takes ideas to physical prototypes — concept, CAD, printing on my own 3D printer, wiring and soldering — then redesigns what fails in testing. Team lead on every team project. Designer, not a programmer: I write the spec, direct AI to write most of the code, and check the results myself.

SELECTED PROJECTS

METO — small interactive robot for learning block coding — Research → thesis · 2025–present

- Research asks whether a robot that shows emotion helps beginners learn block coding differently from one that does not. Led a 3-person thesis team; the planned study will compare it with a robot that does not show emotion among first-year KMUTT students (ethics documents in preparation). The camera reads facial expressions; the robot turns to follow and mirrors them on its screen.
- It has two arms (tested on the Apr 2026 version) and runs wirelessly on two rechargeable 18650 cells with a protection board (BMS) and USB-C charging.
- "METO Robot" web: designed its screens and 62-piece avatar; one account, with teacher/student roles set by class; teacher-set tasks count only when code runs error-free; AI gives hints, not answers.
- Body & mechanism: trapezoid body so the arms can close in and grip; five stacked layers like a tiffin carrier since I have no heat-set inserts; printed oval arms with anti-slip bumps lifted a cup; wheels → printed tracks → two wheels + caster (planned).

Magnetic Universe — magnetic LED wall panel — Jan – May 2026 · course project · team lead

- My idea: one ring of light per magnet (up to two) that floats between sensors with the field strength. 400 × 400 mm panel with 16 Hall sensors (4 × 4); argued against averaging all 16, so position is computed from the strongest sensor and its neighbours, weighted by field strength.
- Designed and printed threaded rods with twist-on caps; caught threads too thin for the slicer to print. The LEDs still showed as dots; packed heat-resistant fibre fill into the sensor layer to diffuse the light, instead of thickening the 6 cm frame, already 3–4 kg and mounted at 60°.

DeepCore → DeepSphere — hands-on deep-sea explorer — 2026 – present · 1st Place · team lead

- Chose a physical device over a game, map or dashboard; 44 real species backed by 10 peer-reviewed sources; no invented animals. DeepCore (hackathon winner): ESP32 controller assembled from real parts, driving a live 3D web scene; I set which sounds we needed and a teammate sourced them.
- DeepSphere is in progress: the six-screen ring is built, but most screens need re-wiring; the three-layer sphere is still the long-term concept. Set sizes and print rules — no supports, thin walls, snap-fits since I don't have heat-set inserts; modelled v1 myself, v2 CAD with AI help.

MOFYCH Commission Website — service design — honeykung.github.io/commission-tag

- Designed the whole site in Figma around an "art + web" idea (hand-drawn clouds and logo, my own animation); tag sign-up with email alerts and email-confirmed cancellation; queue sheet on Google Apps Script. Built with AI from my design.
- Paint matcher for exact colours: transcribed most brand colour charts myself and checked the rest against published data; required one search across all 5 brands (1,131 paints) to get the truest match; set a ΔE < 5 target; dropped fluorescent/metallic paints from 2-paint mix recipes. Matching uses standard CIEDE2000 formulas written by AI.

Smaller builds — self-initiated

- **Minecraft Pixel Lamp** — designed in Tinkercad, printed myself; ESP32 + LEDs. Brightness and current are capped in code (5 V, 1.2 A) so it runs from a phone charger; specified a flame that steps yellow → orange → red from bottom to top. Size: 10 × 10 × 15.4 cm.
- **svgtostl** — tool that turns drawings into printable models; I use it for keychains and 3D-printing sewing-pattern templates: I lay the printed piece on fabric, trace its edge and cut. Set the standard (sharp corners, curves true to the drawing) and rejected point reduction because it turned curves into visible facets for 0.2 mm-nozzle printing.
- **Cotton dolls** — on purchased/public patterns: my design sheets, embroidered faces, removable magnetic ears; OC-only, as purchased patterns bar licensed characters; developing my own pattern in Blender.
- **OC Companion** (in design) — a palm-sized device (about 4–5 cm per side) with a small AMOLED screen as its whole face and figure-style sculpted hair; it sits on a desk, travels with you, and will be made as a keychain, with one device made to order for each buyer's own character (OC).
- **Mini paint palette** — 18 wells in a 109 × 75 mm PLA tray (most small palettes have 5–10), with a side channel for test mixing; in use.
- **MOFYCH collector cards** — 4 designs × 4 cards, 90 × 50 mm, artwork split into layers for holographic foil and matte/smooth finishes, printed by a factory.